ham10000_split

August 10, 2026

[1]:
```
import os
import pandas as pd
import numpy as np
```

[2]:
```
TRAIN_RATIO = 0.7
VAL_RATIO = 0.1
TEST_RATIO = 0.2
RANDOM_SEED = 123
```

[3]:
```
rng = np.random.default_rng(RANDOM_SEED)
```

[4]:
```
assert abs(TRAIN_RATIO + VAL_RATIO + TEST_RATIO - 1.0) < 1e-6
```

[5]:
```
df = pd.read_csv("HAM10000_metadata_official.csv")
```

[6]:
```
df.drop(columns=["dx_type", "dataset"], inplace=True)
```

[7]:
```
df["dataset"] = "ham10000"
```

[8]:
```
df.rename(columns={
    "image_id": "image_name",
    "dx": "label",
    "lesion_id": "patient_id",
    "localization": "location",
}, inplace=True)
```

[9]:
```
df["age"] = df["age"].fillna("unknown")
```

[10]:
```
df.isna().sum()
```

[10]:
```
patient_id    0
image_name    0
label         0
age           0
sex           0
location      0
dataset       0
dtype: int64
```

[11]:
```
assert df.isnull().sum().sum() == 0
```

[12]:
```
df["image_name"] = df["image_name"].apply(
    lambda x: x if str(x).endswith(".jpg") else str(x) + ".jpg")
```

[13]:
```
assert len(df) == df["image_name"].nunique()
```

[14]:
```
missing = []
for f in df["image_name"]:
    if not os.path.exists(os.path.join("/home/prasad/datasets/ham10000/images/
 ↪", f)):
        missing.append(f)

print(len(missing))
```

```
0
```

[15]:
```
mapping = {
    "akiec": "AKIEC",
    "bcc"  : "BCC",
    "bkl"  : "BKL",
    "df"   : "DF",
    "mel"  : "MEL",
    "nv"   : "NEV",
    "vasc" : "VASC",
}
```

[16]:
```
assert set(df["label"].unique()) == set(mapping.keys())
```

[17]:
```
df["label"] = df["label"].map(mapping)
```

[18]:
```
assert df["label"].isna().sum() == 0
```

[19]:
```
assert sorted(df["label"].unique()) == [
    "AKIEC", "BCC", "BKL", "DF", "MEL", "NEV", "VASC"]
```

[20]:
```
df["label"].value_counts()
```

[20]:
```
label
NEV      6705
MEL      1113
BKL      1099
BCC       514
AKIEC     327
VASC      142
DF        115
Name: count, dtype: int64
```

[21]:
```
print(df["label"].value_counts().sum())
```

```
10015
```

[22]:
```
assert df["label"].isna().sum() == 0
```

[23]:
```
df["binary_label"]=(df["label"].isin(
    ["MEL", "BCC", "AKIEC"])).astype(int)
```

[24]:
```
assert set(df["binary_label"].unique()) == {0,1}
```

[25]:
```
print(df["binary_label"].value_counts())
```

```
binary_label
0    8061
1    1954
Name: count, dtype: int64
```

[26]:
```
print(df["binary_label"].value_counts().sum())
```

```
10015
```

[27]:
```
print(pd.crosstab(df["binary_label"], df["label"]))
```

```
label         AKIEC  BCC   BKL   DF   MEL   NEV  VASC
binary_label
0                 0    0  1099  115     0  6705   142
1               327  514     0    0  1113     0     0
```

[28]:
```
df["sex"].value_counts()
```

[28]:
```
sex
male       5406
female     4552
unknown      57
Name: count, dtype: int64
```

[29]:
```
print(df["sex"].value_counts().sum())
```

```
10015
```

[30]:
```
print(pd.crosstab(df["sex"], df["label"]))
```

```
label    AKIEC  BCC  BKL  DF  MEL   NEV  VASC
sex
female     106  197  463  52  424  3237    73
male       221  317  626  63  689  3421    69
unknown      0    0   10   0    0    47     0
```

[31]:
```
df["age"].value_counts()
```

[31]:
```
age
45.0       1299
50.0       1187
55.0       1009
40.0        985
60.0        803
70.0        756
35.0        753
65.0        731
75.0        618
30.0        464
80.0        404
85.0        290
25.0        247
20.0        169
5.0          86
15.0         77
unknown      57
10.0         41
0.0          39
Name: count, dtype: int64
```

[32]:
```
df["age"].value_counts().sum()
```

[32]:
```
np.int64(10015)
```

[33]:
```
df["age"].isna().sum()
```

[33]:
```
np.int64(0)
```

[34]:
```
print(pd.crosstab(df["age"], df["label"]))
```

```
label    AKIEC  BCC  BKL  DF  MEL   NEV  VASC
age
0.0          0    0    5   0    0    30     4
5.0          0    0    1   0    1    81     3
10.0         0    0    0   0    0    39     2
15.0         0    0    0   0    0    73     4
20.0         0    1    0   0    6   158     4
25.0         0    3    0   2   16   221     5
30.0         1    4    6   4   34   410     5
35.0         0    5   24  12   36   668     8
40.0         9   23   46   9   49   846     3
45.0        10   26   59  14   74  1100    16
50.0        19   27   87  18   96   928    12
55.0        27   25   95  13  142   686    21
60.0        58   35  131   9  106   454    10
65.0        38   79  108  18  133   351     4
```

```
70.0        56   85  183   4  166   248    14
75.0        47   76  153   9   91   231    11
80.0        37   73   98   3   85    97    11
85.0        25   52   93   0   76    39     5
unknown      0    0   10   0    2    45     0
```

[35]:
```
df["location"].value_counts()
```

[35]:
```
location
back               2192
lower extremity    2077
trunk              1404
upper extremity    1118
abdomen            1022
face                745
chest               407
foot                319
unknown             234
neck                168
scalp               128
hand                 90
ear                  56
genital              48
acral                 7
Name: count, dtype: int64
```

[36]:
```
df["location"].value_counts().sum()
```

[36]:
```
np.int64(10015)
```

[37]:
```
df["location"].isna().sum()
```

[37]:
```
np.int64(0)
```

[38]:
```
print(pd.crosstab(df["location"], df["label"]))
```

```
label            AKIEC  BCC  BKL  DF  MEL   NEV  VASC
location
abdomen              5   18   42   4   66   860    27
acral                0    0    0   0    0     7     0
back                29  186  202   2  324  1427    22
chest               12   47   74   0   68   200     6
ear                  3    0    6   0   17    30     0
face               113  101  319   0  104   100     8
foot                 0    4    1   2   28   284     0
genital              0    0    2   1    0    45     0
hand                13    2   14   0    1    55     5
lower extremity     65   58  174  82  192  1479    27
neck                10   14   31   0   29    81     3
```

```
scalp               14   19   34   0   14    45     2
trunk                1   11   73   0   47  1241    31
unknown              0    5   23   0   10   196     0
upper extremity     62   49  104  24  213   655    11
```

1 Split

[39]:
```
patient_ids = df["patient_id"].dropna().unique().to_numpy()
```

[40]:
```
rng.shuffle(patient_ids)
```

[41]:
```
df["patient_id"].isna().sum()
```

[41]:
```
np.int64(0)
```

[42]:
```
n_patients = len(patient_ids)
n_train = int(TRAIN_RATIO * n_patients)
n_val = int(VAL_RATIO * n_patients)
```

[43]:
```
train_patients = patient_ids[:n_train]
val_patients = patient_ids[n_train:n_train + n_val]
test_patients = patient_ids[n_train + n_val:]
```

[44]:
```
train_ids = set(train_patients)
val_ids   = set(val_patients)
test_ids  = set(test_patients)

print("Train  Val :", train_ids & val_ids)
print("Train  Test:", train_ids & test_ids)
print("Val  Test  :", val_ids & test_ids)
```

```
Train  Val : set()
Train  Test: set()
Val  Test  : set()
```

[45]:
```
df["split"] = "unassigned"
df.loc[df["patient_id"].isin(train_patients), "split"] = "train"
df.loc[df["patient_id"].isin(val_patients), "split"] = "val"
df.loc[df["patient_id"].isin(test_patients), "split"] = "test"
```

[46]:
```
assert (df["split"] != "unassigned").all()
```

[47]:
```
print("Patients per split:")
print(df.groupby("split")["patient_id"].nunique())
```

```
Patients per split:
split
test     1494
```

```
train    5229
val       747
Name: patient_id, dtype: int64
```

[48]:
```
print("\nImages per split:")
print(df["split"].value_counts())
```

```
Images per split:
split
train    7034
test     1982
val       999
Name: count, dtype: int64
```

[49]:
```
print("\nClass distribution per split:")
print(pd.crosstab(df["split"], df["label"]))
```

```
Class distribution per split:
label  AKIEC  BCC  BKL  DF  MEL   NEV  VASC
split
test      72  109  194  24  220  1342    21
train    218  357  783  82  788  4699   107
val       37   48  122   9  105   664    14
```

[50]:
```
print(df.groupby(["split", "label"]).size().unstack(fill_value=0))
```

```
label  AKIEC  BCC  BKL  DF  MEL   NEV  VASC
split
test      72  109  194  24  220  1342    21
train    218  357  783  82  788  4699   107
val       37   48  122   9  105   664    14
```

[51]:
```
pd.crosstab(df["age"], df["split"])
```

[51]:

split	test	train	val
age			
0.0	8	31	0
5.0	17	64	5
10.0	13	24	4
15.0	11	56	10
20.0	29	115	25
25.0	52	160	35
30.0	90	325	49
35.0	139	529	85
40.0	190	704	91
45.0	259	928	112
50.0	244	834	109

```
55.0      182    701  126
60.0      167    567   69
65.0      147    530   54
70.0      161    506   89
75.0      131    437   50
80.0       65    288   51
85.0       64    198   28
unknown    13     37    7
```

[52]:
```
pd.crosstab(df["sex"], df["split"])
```

[52]:
```
split    test  train  val
sex
female    906   3169  477
male     1066   3825  515
unknown    10     40    7
```

[53]:
```
pd.crosstab(df["location"], df["split"])
```

[53]:
```
split            test  train  val
location
abdomen           216    712   94
acral               0      7    0
back              410   1571  211
chest              98    268   41
ear                 8     41    7
face              143    530   72
foot               78    211   30
genital            14     32    2
hand               22     60    8
lower extremity   418   1442  217
neck               32    120   16
scalp              24     92   12
trunk             273    996  135
unknown            48    156   30
upper extremity   198    796  124
```

[54]:
```
df.groupby(["split", "sex", "label"]).size()
```

[54]:
```
split  sex      label
test   female   AKIEC      29
                BCC        50
                BKL        84
                DF          9
                MEL        83
                NEV       638
                VASC       13
```

```
        male     AKIEC      43
                 BCC        59
                 BKL       107
                 DF         15
                 MEL       137
                 NEV       697
                 VASC        8
        unknown  BKL         3
                 NEV         7
train   female   AKIEC      61
                 BCC       132
                 BKL       314
                 DF         40
                 MEL       305
                 NEV      2267
                 VASC       50
        male     AKIEC     157
                 BCC       225
                 BKL       464
                 DF         42
                 MEL       483
                 NEV      2397
                 VASC       57
        unknown  BKL         5
                 NEV        35
val     female   AKIEC      16
                 BCC        15
                 BKL        65
                 DF          3
                 MEL        36
                 NEV       332
                 VASC       10
        male     AKIEC      21
                 BCC        33
                 BKL        55
                 DF          6
                 MEL        69
                 NEV       327
                 VASC        4
        unknown  BKL         2
                 NEV         5
dtype: int64
```

[55]:
```python
assert df["image_name"].is_unique
assert df["split"].isin(["train", "val", "test"]).all()
assert df["sex"].isin(["male", "female", "unknown"]).all()
assert df["label"].notna().all()
```

[56]:
```
leakage = (
    df.groupby("patient_id")["split"]
    .nunique()
    .gt(1)
    .sum()
)
print(f"Patients appearing in multiple splits: {leakage}")
```

```
Patients appearing in multiple splits: 0
```

[57]:
```
df=df[["dataset", "split", "image_name", "label", "binary_label", "patient_id",
       "age", "sex", "location"]]
```

[58]:
```
expected_cols = [
    "dataset",
    "split",
    "image_name",
    "label",
    "binary_label",
    "patient_id",
    "age",
    "sex",
    "location",
]
assert list(df.columns) == expected_cols
```

[59]:
```
print("=" * 60)
print("HAM10000 metadata preprocessing completed")
print(f"Images   : {len(df)}")
print(f"Patients : {df['patient_id'].nunique()}")
print(f"Classes  : {df['label'].nunique()}")
print(df["split"].value_counts())
print("=" * 60)
```

```
============================================================
HAM10000 metadata preprocessing completed
Images   : 10015
Patients : 7470
Classes  : 7
split
train    7034
test     1982
val       999
Name: count, dtype: int64
============================================================
```

[60]:
```
print(df.isnull().sum())
```

```
dataset         0
```

```
split           0
image_name      0
label           0
binary_label    0
patient_id      0
age             0
sex             0
location        0
dtype: int64
```

[61]:
```python
assert df["image_name"].is_unique
assert df["patient_id"].notna().all()
assert df["label"].notna().all()
assert df["binary_label"].isin([0, 1]).all()
assert df["split"].isin(["train", "val", "test"]).all()
```

[62]:
```python
assert df.isnull().sum().sum() == 0
```

[63]:
```python
df.to_csv("ham10000_metadata_clean.csv", index=False, encoding="utf-8")
```

[64]:
```python
df
```

[64]:

	dataset	split	image_name	label	binary_label	patient_id	age	sex	location
0	ham10000	train	ISIC_0027419.jpg	BKL	0	HAM_0000118	80.0	male	scalp
1	ham10000	train	ISIC_0025030.jpg	BKL	0	HAM_0000118	80.0	male	scalp
2	ham10000	val	ISIC_0026769.jpg	BKL	0	HAM_0002730	80.0	male	scalp
3	ham10000	val	ISIC_0025661.jpg	BKL	0	HAM_0002730	80.0	male	scalp
4	ham10000	train	ISIC_0031633.jpg	BKL	0	HAM_0001466	75.0	male	ear
…	…	…	…	…	…	…	…	…	…
10010	ham10000	test	ISIC_0033084.jpg	AKIEC	1	HAM_0002867	40.0	male	abdomen
10011	ham10000	test	ISIC_0033550.jpg	AKIEC	1	HAM_0002867	40.0	male	abdomen
10012	ham10000	test	ISIC_0033536.jpg	AKIEC	1	HAM_0002867	40.0	male	abdomen
10013	ham10000	train	ISIC_0032854.jpg	AKIEC	1	HAM_0000239	80.0	male	face
10014	ham10000	train	ISIC_0032258.jpg	MEL	1	HAM_0003521	70.0	female	back

```
[10015 rows x 9 columns]
```